UNIVERSITY OF CAMBRIDGE

Engineering Tripos Part IIA Extension Activity (ExA):

Wireless Communication with Software-Defined Radio (SDR)

Open to all engineering students, but particularly recommended for those in Electrical and Electronic Engineering, Information and Computer Engineering, or Electrical and Information Sciences.

Introduction to OFDM and SDR for Wireless Communication

In this ExA, you will explore how digital information can be transmitted through a wireless channel using Orthogonal Frequency Division Multiplexing, known as OFDM. OFDM is used in many modern communication systems, including Wi-Fi, 4G, 5G, and digital broadcasting, because it allows data to be transmitted efficiently over many closely spaced frequency components.

The activity begins with signal processing concepts that you have already met, especially sampling and the Fast Fourier Transform (FFT). You will use Python to build a simple OFDM transmitter and receiver step by step. First, digital bits will be represented by simple numerical symbols. These symbols will then be placed onto different frequency positions, called subcarriers. An inverse FFT will combine the subcarriers into one time-domain waveform for transmission. At the receiver, an FFT will separate the waveform back into its subcarriers so that the original symbols, and then the original bits, can be recovered.

After developing the simulation, you will use the ADALM-Pluto Software Defined Radio (SDR) platform to transmit and receive signals in a real wireless environment. This will allow you to compare ideal simulation results with practical measurements. You will observe how effects such as noise, timing error, multipath propagation, gain settings, and frequency differences between transmitter and receiver can change the received signal.

The aim of this activity is to develop a practical understanding of how OFDM works and why it is useful in wireless communication. By building the system from first principles and testing it using SDR hardware, you will connect FFT-based signal processing with real over-the-air transmission.

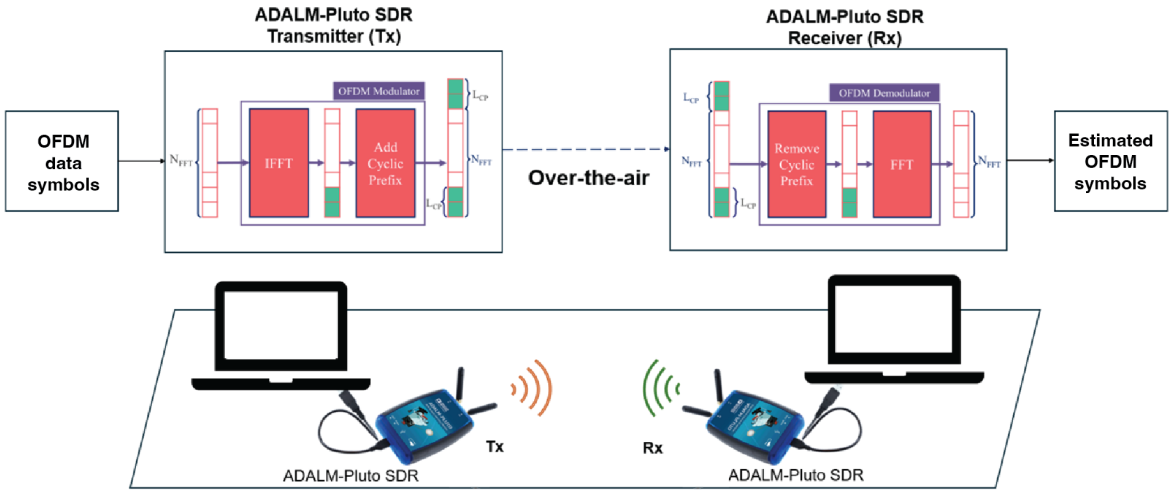


Figure 1: Two-Device Setup for Real-Time OFDM Transmission.

Objectives

By completing this activity, you will develop a practical understanding of how OFDM-based wireless communication works, from simulation to real SDR transmission. You will:

- use the FFT to analyse signals in the frequency domain
- explain how digital bits can be represented using simple modulation symbols such as BPSK or QPSK
- build a basic OFDM transmitter and receiver using the IFFT and FFT
- investigate how noise, multipath, timing errors, and frequency offset affect received signals
- explain the purpose of a cyclic prefix in reducing the effect of delayed signal paths
- use ADALM-Pluto SDR to transmit, receive, and observe simple wireless signals
- compare simulation results with real SDR measurements
- interpret time-domain plots, spectra, constellation diagrams, and bit-error results
- present your implementation, results, observations, and conclusions clearly

Part 1: Simulation-Based OFDM

In the first part of the activity, you will simulate a simple communication system.

Task 1: Signals and FFT Review

You will begin by generating simple sinusoidal signals and observing them in the time and frequency domains. If the sampling rate is F_S samples per second and there are (N) samples, the time values can be written as:

$$t = \frac{0,1,2,\ldots,N-1}{F_s}$$

A sinusoidal signal can be generated using $x(t) = A\cos(2\pi ft)$, where *A* is the amplitude and *f* is the frequency in Hz.

You will use FFT to convert a sampled time-domain signal into its frequency-domain representation. You should investigate:

- a single sinusoidal signal
- the sum of two or more sinusoids
- how the FFT shows peaks at the frequencies present in the signal
- how the sampling rate (F_s) changes the maximum observable frequency
- how the number of samples (N) changes the frequency resolution

Task 2: Digital Symbols

You will generate a random sequence of bits and map them to simple digital modulation symbols. A modulation symbol is a numerical value used to represent one or more bits before transmission. For BPSK (Binary Phase Shift Keying), each bit is mapped to one symbol:

- bit 0 → symbol (-1)
- bit 1 → symbol (+1)

This is the simplest modulation. For QPSK (Quadrature Phase Shift Keying), two bits are mapped to one complex symbol (in-phase and quadrature components). A simple mapping is:

- bits 00 → $\left((1+j)/\sqrt{2}\right)$
- bits 01 → $\left((-1+j)/\sqrt{2}\right)$
- bits 11 → $\left((-1-j)/\sqrt{2}\right)$
- bits 10 → $\left((1-j)/\sqrt{2}\right)$

The division by $(\sqrt{2})$ keeps the symbol power normalised.

A constellation diagram is a plot of the transmitted or received symbols on the complex plane. The horizontal axis shows the real part, and the vertical axis shows the imaginary part. For BPSK, there are two points. For QPSK, there are four points.

At the receiver, demodulation is done by deciding which constellation point the received symbol is closest to. For BPSK, this can be done simply: received value greater than 0 → bit 1, received value less than 0 → bit 0. For QPSK, the sign of the real part gives one bit and the sign of the imaginary part the other.

Task 3: Building a Simple OFDM Symbol

You will now create a basic OFDM symbol. OFDM sends data using many closely spaced frequency components called subcarriers. Each subcarrier carries one modulation symbol, such as a BPSK or QPSK symbol. In a simulation, subcarriers are represented by FFT bins. An FFT bin is one position in the frequency-domain array. For example, if the FFT size is 64, there are 64 frequency bins.

At the transmitter, you will create a frequency-domain array and place modulation symbols into selected bins. These selected bins are called active subcarriers. For the first implementation, use FFT size N = 64 and place BPSK (then QPSK) symbols into 16 active subcarriers. For example, use subcarrier indices 1 to 16 in both the transmitter and receiver, and set all other bins to zero. The receiver must extract the same bins after applying the FFT.

The IFFT (Inverse Fast Fourier Transform) converts the frequency-domain subcarrier values into a time-domain OFDM signal. This time-domain signal is what would be transmitted.

At the transmitter:

1. Generate random bits.
2. Map the bits to BPSK or QPSK symbols.
3. Create a frequency-domain array of length (N).
4. Place the modulation symbols onto selected FFT bins.
5. Set unused bins to zero.
6. Apply the IFFT to produce the time-domain OFDM symbol.

At the receiver:

1. Take the received time-domain OFDM symbol.
2. Apply the FFT.
3. Extract the same active subcarriers used by the transmitter.
4. Recover the modulation symbols.
5. Demodulate the symbols back into bits.

In an ideal channel with no noise or distortion, the received bits should exactly match the transmitted bits. This task shows the key idea of OFDM: the IFFT combines many subcarriers into one time-domain waveform, and the FFT separates them again at the receiver.

Task 4: Adding Noise

You will add noise to the transmitted OFDM signal and observe how the receiver performance changes. Additive White Gaussian Noise, or AWGN, is a common model for random noise in communication systems. “Additive” means the noise is added to the signal. “White” means it affects all frequencies equally. In code, noisy received samples can be generated as:

$$r[n] = x[n] + w[n]$$

Where $x[n]$ is the transmitted OFDM sample and $w[n]$ is random noise. The Signal-to-Noise Ratio (SNR) compares the signal power to the noise power. A high SNR means the signal is much stronger than the noise. A low SNR means the noise is large compared with the signal. For a chosen SNR in dB, the noise power can be calculated using:

$$P_{\text{noise}} = \frac{P_{\text{signal}}}{10^{\text{SNR}_{dB}/10}}$$

For complex baseband OFDM signals, the noise can be generated using independent real and imaginary Gaussian noise samples. You should investigate:

- low-noise and high-noise cases
- constellation diagrams before and after adding noise
- how noise moves received symbols away from their ideal constellation points
- how noise causes bit errors

The Bit Error Rate (BER) is:

$$BER = \frac{\text{number of incorrectly received bits}}{\text{total number of transmitted bits}}$$

To measure BER reliably, repeat the transmission for many random OFDM symbols and count the total number of bit errors over all symbols. You should plot or record how BER changes as SNR is varied. The expected result is that BER decreases as SNR increases.

Task 5: Simple Multipath

You will investigate how OFDM behaves when the received signal contains delayed copies of the transmitted signal. Multipath occurs when a wireless signal reaches the receiver through more than one path. For example, part of the signal may travel directly from transmitter to receiver, while another part may reflect from a wall and arrive slightly later.

A simple multipath channel can be simulated using a short impulse response. For example, $h = [1, 0, 0.4]$ means the receiver gets the original signal plus a copy delayed by two samples and reduced to 40% strength. In code, the received signal is generated by convolving the transmitted signal with the channel ($r = x * h$). A mild echo like this mainly distorts the constellation. To see clear bit errors, also try a stronger, longer echo such as h = [1, 0, 0, 0, 0, 0.7].

Multipath can cause one OFDM symbol to interfere with another. It can also disturb the subcarriers when the receiver applies the FFT. A cyclic prefix is used to reduce this problem. It is created by copying the last part of the OFDM symbol and adding it to the beginning. For example, if the OFDM symbol has 64 samples and the cyclic prefix length is 16 samples:

- Take the last 16 samples of the OFDM symbol.
- Place them at the front.
- Transmit the longer symbol of 80 samples.

At the receiver:

- Remove the first 16 samples.
- Apply the FFT to the remaining 64 samples.

The cyclic prefix protects the useful OFDM symbol from delayed copies of the previous symbol. If the cyclic prefix is longer than the channel delay, removing the prefix leaves a clean block of N samples, and the channel acts as a simple multiply on each subcarrier. Each subcarrier is then scaled and rotated by the channel response H[k] = FFT(h), so you must divide each recovered subcarrier by H[k] (a one-tap equaliser) to restore the symbols. H can be assumed known or estimated from a known reference symbol.

You should compare OFDM performance with and without a cyclic prefix, looking at both the received constellation and the BER. Note that for a mild echo the damage may appear only as constellation spread rather than bit errors, while a stronger echo produces clear errors without a cyclic prefix. In both cases a sufficient cyclic prefix should restore a clean constellation and a BER of zero in the absence of noise.

Task 6: Simple Frequency Offset Observation

You will introduce a small frequency offset and observe its effect on the received OFDM signal.

A frequency offset occurs when the transmitter and receiver are not tuned to the same carrier frequency. This is common in real wireless hardware because oscillators are not perfectly matched. In simulation, frequency offset can be added by multiplying the transmitted samples by a rotating complex exponential:

$$r[n] = x[n]e^{j2\pi f_{\text{offset}} n/F_s}$$

Where $x[n]$ is the transmitted sample; f_{offset} is the frequency offset in Hz; F_s is the sampling rate; n is the sample index.

A small frequency offset may cause the received constellation to rotate. A larger offset can also cause energy from one subcarrier to leak into other subcarriers. This is one reason OFDM systems need good synchronisation. For the main activity, you only need to observe and explain the effect. As an optional extension, you may try a simple correction by multiplying the received signal by the opposite rotation:

$$r_{\text{corrected}}[n] = r[n]e^{-j2\pi f_{\text{offset}} n/F_s}$$

This correction only works if the frequency offset is known or estimated.

Part 2: SDR-Based Wireless Experiments

In the second part of the activity, you will use the ADALM-Pluto SDR platform to transmit and receive a real OFDM signal over the air and compare what you measure against your Part 1 simulation.

In SDR experiments, signals are represented as IQ samples. An IQ sample is a complex number whose real part is the in-phase component and whose imaginary part is the quadrature component. The centre (carrier) frequency sets the radio frequency used for transmission and reception. The sampling rate sets how many IQ samples are processed per second; the subcarrier spacing and FFT size together fix this rate. Transmit and receive gain control the signal strength at the two ends.

You will use two ADALM-Pluto radios on two host computers (one transmitting, one receiving). The transmitter and receiver scripts share an identical set of parameters. Installing the drivers and libraries, connecting the radio, finding its device address (URI), and verifying the connection are platform-specific and are documented in the accompanying ADALM-Pluto Setup Guide.

Task 7: Setup and Connection Check

Before transmitting OFDM, confirm the hardware is configured and reachable.

1. Following the setup guide, confirm each radio connects and reports its sample rate from your code.
2. Set the same carrier frequency, sampling rate, and bandwidth on both radios. A 2.4 GHz carrier works but shares the band with Wi-Fi/Bluetooth, a quieter band (e.g. 915 MHz) is often cleaner.
3. As an initial sanity check, make the link as easy as possible: place the two radios close together. A strong, clean link lets you confirm the full chain works before you introduce distance and interference.

Task 8: Transmitting and Receiving an OFDM Waveform

In this task you will run a complete OFDM link over the air: one packet is generated, transmitted repeatedly by the SDR, captured at the receiver, and decoded. Most of the Part 2 code is provided in the lab: the transmitter, the receiver, and the detection, frequency-offset, channel-estimation, and phase-tracking algorithms. Your job is to run the system, explore how it behaves, and interpret what each stage does, not to write it from scratch. The descriptions below follow the signal through the system so that you can explain your results.

Packet structure: The packet begins with a known preamble, followed by several OFDM data symbols, each protected by a cyclic prefix. The preamble is one known OFDM symbol transmitted twice (two identical halves); this single structure serves three purposes at the receiver: it locates the start of the packet, it provides the carrier-frequency-offset estimate, and it provides the channel estimate. Unlike the minimal Part 1 example (16 subcarriers on bins 1–16), the SDR system spreads its subcarriers on both sides of the DC (zero-frequency) subcarrier and leaves the DC subcarrier and the band edges empty: real hardware leaks energy at DC, and the edge subcarriers are kept as guard bands that absorb the transmit/receive filter roll-off. Within each data symbol, a few subcarriers carry pilots (known reference values used to track residual phase).

Payload: By default, the payload is a short text message, a small image, or both, packed into one bitstream. The bitstream begins with a fixed-size length header so the receiver can separate the payload from padding and reconstruct the exact contents. The image is sent as a raw bitstream of its pixel bytes and reshaped back into an image at the receiver. BPSK is recommended for the first working link because it is the most robust to real-channel impairments; QPSK doubles the data rate but is more sensitive (see Task 9).

What the transmitter does:

1. Build the payload bitstream: length header, then the text bytes, then the image bytes.
2. Map the bits to BPSK or QPSK symbols (as in Part 1).

3. Spread the symbols across the data subcarriers of successive OFDM symbols, inserting the known pilot values.
4. Apply the IFFT to each symbol and add the cyclic prefix.
5. Prepend the preamble, normalise the whole packet, and scale it for the radio's digital-to-analogue converter.
6. Transmit the packet continuously (cyclic buffer).

What the receiver does:

1. Capture a buffer of IQ samples long enough to contain at least one complete packet.
2. Detect the start of the packet using the known preamble.
3. Estimate and correct the carrier frequency offset.
4. Remove the cyclic prefix from each data symbol and apply the FFT.
5. Estimate the channel from the preamble and equalise each subcarrier.
6. Track residual phase per symbol using the pilots.
7. Extract the data subcarriers, demodulate to bits, read the length header, and recover the text and image.
8. Where the transmitted bits are known, compute the bit error rate (BER).

The effects the receiver must handle: Steps 2, 3, 5 and 6 exist to undo three things the real channel and hardware do to the signal. You should understand these conceptually.

Packet timing: The receiver must find where the packet begins within the captured samples. Because the preamble is two identical halves, the receiver recognises it by looking for this repetition, which produces a strong and easily detected response even when a frequency offset is present. The detected position marks the start of the packet.

Carrier frequency offset: The transmitter and receiver oscillators are never perfectly matched, so the received signal carries a frequency offset, which affects OFDM in two very different ways:

- It rotates the whole constellation, with the rotation growing over the duration of the packet. This part can largely be tracked and removed using the pilots.
- More seriously, when the offset becomes a significant fraction of the subcarrier spacing, it breaks the orthogonality between subcarriers (the subcarriers are spaced so that each sits exactly on a null of all the others), so energy leaks from each subcarrier into its neighbours (inter-carrier interference). The pilots cannot remove this; it can only be avoided by estimating and correcting the offset itself.

For this reason, the receiver first estimates the offset from the repeated preamble and counter-rotates the signal, and only then relies on the pilots for any small residual phase. This is why OFDM systems need good synchronisation.

Channel effects: The wireless channel scales and rotates each subcarrier differently (frequency-selective fading from multipath). The receiver estimates this from the known preamble and equalises each subcarrier. A small timing offset additionally appears as a phase that varies linearly across subcarriers, which the equaliser absorbs.

Produce and keep for your presentation:

- the captured |IQ| with the detected packet start marked;
- the detection metric across the buffer;
- the received spectrum of a data symbol;
- the channel estimate (magnitude and phase across subcarriers);
- the received constellation before and after frequency-offset correction, so the effect of the correction is visible;
- the recovered text, the reconstructed image, and the measured BER.

Once you have a working link, vary the transmit gain, the receive gain, and the distance between the radios, and record how the received magnitude, spectrum, constellation, and BER change. Note in particular that too little receive gain buries the signal in noise, while too much causes clipping (saturation). Keep these captures for the comparison in Task 9.

Task 9: Comparing Simulation and Hardware Results

Compare your Part 1 simulation with your SDR measurements. You should discuss, supported by your plots:

- why the simulated system is cleaner than the real one (ideal channel and perfect synchronisation in simulation versus noise, hardware impairments, and imperfect timing/frequency alignment in hardware)
- how noise and the gain settings affect the received constellation and the BER, using the transmit-gain, receive-gain, and distance captures from Task 8
- the effect of the frequency offset, using your before-and-after-correction constellations (see the Carrier frequency offset notes above); report your measured offset as a fraction of the subcarrier spacing
- why the received constellation may still rotate or spread after correction (residual offset, phase noise, and the frequency-selective multipath channel)
- the difference between BPSK and QPSK over the air: compare the BER and the constellation you obtain for each, and explain why BPSK tolerates rotation and spread better
- why packet detection may not always succeed (low gain, distance, interference), and what your detection-metric plot shows

Optional Extension Tasks

Students who make strong progress may attempt one or more of the following:

- test the system at several distances and relate the results to your channel estimate
- compare different FFT sizes and different cyclic-prefix lengths
- measure BER for a range of receive-gain settings
- investigate how movement near the antennas changes the received signal and the channel estimate over repeated captures

Assessment

Assessment will be based on a 10–15-minute final presentation. Your presentation should include:

- a description of how your simulation transmitter and receiver work

- plots showing time-domain signals, spectra, and constellation diagrams
- results showing the effect of noise and simple channel impairments
- a demonstration or discussion of SDR transmission and reception
- a comparison between simulation and real SDR observations
- conclusions explaining what you learned about OFDM and wireless communication

The total expected time for the activity is approximately 16 hours. Satisfactory completion of the core simulation and SDR tasks will lead to full standard credit.